

Simple 12V To 24V VOLTAGE DOUBLER

SURESH CHANDRA DWIVEDI

Many electronic circuits require a voltage higher than that available from the power supply. For example, when only a 12V battery or adaptor is available but a circuit requires 20V to 24V, a voltage-boosting arrangement becomes necessary. One simple and economical solution is a charge-pump voltage doubler driven by an oscillator.

The system is suitable for small power supplies, op-amp circuits, analogue electronics experiments, and various DIY applications that require a moderately higher DC voltage from a lower-voltage source. Fig. 1 shows the author's prototype.

Circuit and working

Fig. 2 shows the circuit diagram of the voltage doubler system. It is built around the NE555 timer (IC1), transistors BD139 (T1) and BD140 (T2), and a few passive components. The system consists of four main sections: the oscillator stage, the driver stage, the voltage-doubler network, and the output filter stage.

IC1 (NE555) is configured as an astable multivibrator to generate a square-wave signal. Resistors R1 (4.7k Ω) and R2 (22k Ω), together with capacitor C2 (0.1 μ F), set the oscillation frequency to approximately 3kHz. Capacitor C1 (0.01 μ F) is connected to the control-voltage pin to improve stability and reduce noise.

The square-wave signal at pin 3 drives the complementary transistor pair comprising T1 (BD139) and T2 (BD140), operating in push-pull mode. This stage improves current-handling capability and enhances the charging and discharging process

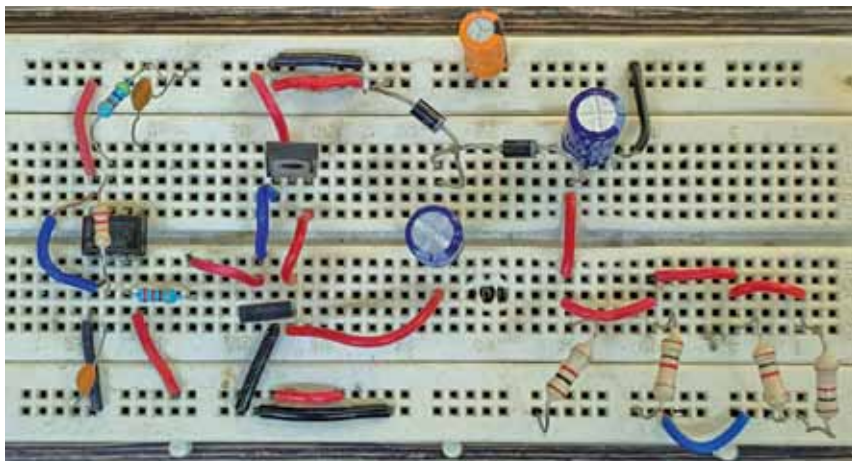


Fig. 1: Author's prototype

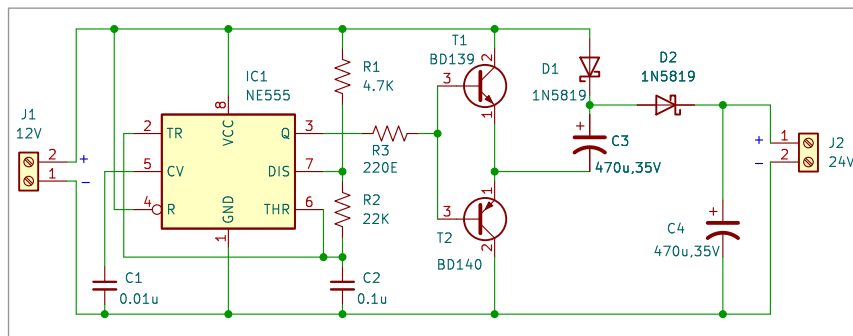


Fig. 2: Circuit diagram

required for efficient voltage doubling.

The voltage-doubler section is formed by Schottky diodes D1 and D2 (both 1N5819), along with capacitors C3 and C4. During one half-cycle of the oscillator output, capacitor C3 charges through diode D1. During the opposite half-cycle, the stored charge in C3 is added to the supply voltage and transferred through diode D2 to capacitor C4. This action produces an output voltage approximately twice the input voltage.

Schottky diodes are selected because of their low forward-voltage drop and fast-switching characteristics, which improve system

efficiency. Capacitor C4 acts as the output filter capacitor, smoothing the pulsating waveform to provide a relatively stable DC output at connector J2.

With a 12V DC supply connected at J1, the system typically produces 20V to 24V DC, depending on the load and component tolerances.

Construction and testing

An actual-size single-sided PCB layout for the voltage doubler system is shown in Fig. 3, while the corresponding component layout is shown in Fig. 4. After assembling the system on the PCB, mount it inside a